Ansh Pachauri

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EDUCATION

The Ohio State University — Columbus, OH

Bachelor of Science - Computer Science and Engineering | Current CGPA: 9.55/10.0 — *Graduation: 10 May 2026*

Specializing in Software Engineering — *Dean's List: 5 semesters*

AWARDS

2nd Place, Fintech Track, IBM SkillsBuild AI Hackathon at OSU (Team X - Solo) — March 2026

- Won $750 solo at IBM SkillsBuild Fintech Hackathon - shipped a production-ready full-stack AI application end-to-end in under 24 hours, outperforming all 100+ participants including every multi-person team.

WORK EXPERIENCE

Product Development and AI Integration Intern — May 2025 - August 2025

NMT Security — *Tinton Falls, NJ*

- Reduced security ticket resolution time by 40% by designing and deploying a full-stack AI chatbot on AWS (Python, Lambda, Bedrock) for internal security teams, autonomously processing 1,000+ daily queries at 95% classification accuracy in production.
- Automated cybersecurity threat intelligence workflows by building Python scripts integrating REST APIs (VirusTotal, OTX) to classify and triage incoming threats, eliminating 15+ hours per week of manual analyst work and improving detection accuracy by 30%.
- Owned the complete SDLC across both deliverables, requirements gathering, system design, coding, testing, and full production deployment on AWS, all delivered within a single 3-month internship with zero additional engineering support from the team.

PROJECTS

Tech Asset & Service Management Portal | *Angular 17, Java 21, Spring Boot 3, PostgreSQL, Google Cloud* — April 2026

- Built a full-stack internal enterprise web application with 4-role RBAC, Employee, Technician, Manager, Admin, enforced at the service layer via Spring Security @PreAuthorize, independently of Angular route guards.
- Implemented a 9-state service request approval workflow and a 6-state asset lifecycle state machine in Spring Boot, enforcing invalid transitions via domain exceptions and writing immutable before/after JSON audit records for every state-changing operation.
- Deployed backend and frontend as separate Cloud Run services on GCP, each continuously deployed from GitHub via Cloud Build, backed by PostgreSQL on Cloud SQL with Flyway migrations, and validated by a GitHub Actions CI pipeline on every push to main.

BuckeyeMealPlanner | *Flask, Gemini 2.5 Flash, IBM Cloud, Docker* — March 2026

- Consumed and normalized 5,900+ live menu items across 34 OSU dining locations via the Nutrislice API, powering a Gemini 2.5 Flash AI engine that auto-generates personalized 7-day meal plans enforcing per-meal calorie, protein, and carb constraints.
- Currently being piloted at The Ohio State University following a formal meeting with the OSU Director of Dining Services to discuss campus-wide implementation, making it the first student-built tool to reach institutional review within OSU Dining Services.
- Deployed on Google Cloud Run via Docker with 200+ automated tests covering API extraction, routing, and edge cases, plus a GitHub Actions daily workflow that automatically re-fetches and commits all updated live menu data every morning at 5:00 AM ET.

E-commerce Platform | *React.js, Node.js, AWS Lambda, DynamoDB, API Gateway* — August 2025 - November 2025

- Designed and built a scalable, reliable full-stack web application load-tested to 500+ concurrent users, decoupled checkout, payment, and inventory into independent AWS Lambda functions via API Gateway with modular, well-structured code and clear contracts.
- Applied user-centered design throughout the front-end, reducing cart abandonment by 25% via an iteratively refined multi-step UX checkout flow, and cut feature iteration time by 40% through strict modular service boundaries and async DynamoDB state updates.
- Implemented the complete front-end to back-end data flow from React/Angular UI through API Gateway into DynamoDB, applying partition key and GSI design patterns to optimize for high-throughput concurrent reads and idempotent write operations at scale.

Real-Time Market Sentiment Engine | Python, Apache Kafka, Docker, FAISS, React — November 2025

- Built a distributed event-driven pipeline using Apache Kafka and Docker that ingests live financial news streams and delivers sub-second bullish/bearish market sentiment signals in real time via retrieval-augmented generation (RAG) with FAISS vector search.
- Designed the complete system architecture, Kafka ingestion layer, Python embedding service, FAISS retrieval engine, and React dashboard, with full horizontal scalability achieved through containerized and independently scalable Kafka consumer groups.
- Implemented RAG with FAISS vector indexing for semantic similarity search across live financial news embeddings, enabling nuanced context-aware market trend detection and directional signal generation that goes well beyond simple keyword-based matching.

LEADERSHIP & INVOLVEMENT

The Quantum Computing Club at Ohio State | *Founder and President* — March 2025 - present

- Launched the university's first quantum computing organization by leading entrepreneurial and outreach efforts, building cross-functional relationships with faculty and student groups, recruiting an executive board, and expanding membership through targeted initiatives.

Student Life | The Ohio State University | *Resident Advisor* — August 2024 - present

- Led community-building for diverse residents, boosting participation by 40% through 15+ events; maintained a 95% satisfaction rate by fostering an inclusive environment and resolving 10+ conflicts with empathetic outreach.

SKILLS

AI-Assisted Development: Prompt Engineering, Workflow Automation, Code Review & Auditing, RAG Implementation.

Languages: Java, Python, JavaScript (Node.js, React, Next.js, TypeScript), Go, Ruby, SQL, C, HTML/CSS

Frameworks & DBs: React.js, Next.js, Node.js, Flask, Ruby on Rails, Tailwind CSS, REST APIs, PostgreSQL, DynamoDB, MySQL, SQLite3

Cloud & DevOps: AWS (Lambda, Bedrock, S3, RDS, EC2, DynamoDB, API Gateway), Docker, Kafka, CI/CD, serverless architecture

CS & ML: Data structures & algorithms, OOP, system design, distributed systems, Agile/Scrum, TDD, NLP, RAG, FAISS

Research: "Fake News Detection using ML & NLP" - IEEE ICTAI 2021, 22 citations, co-authored at age 16